

FundFlow — Fund Flow Dynamics

This diagram set explains how money moves through FundFlow: **two parallel pools** (cash and fund), kept in sync between master and member accounts, with **bank clearance** as a separate reconciliation step.

1. Account map — where money lives

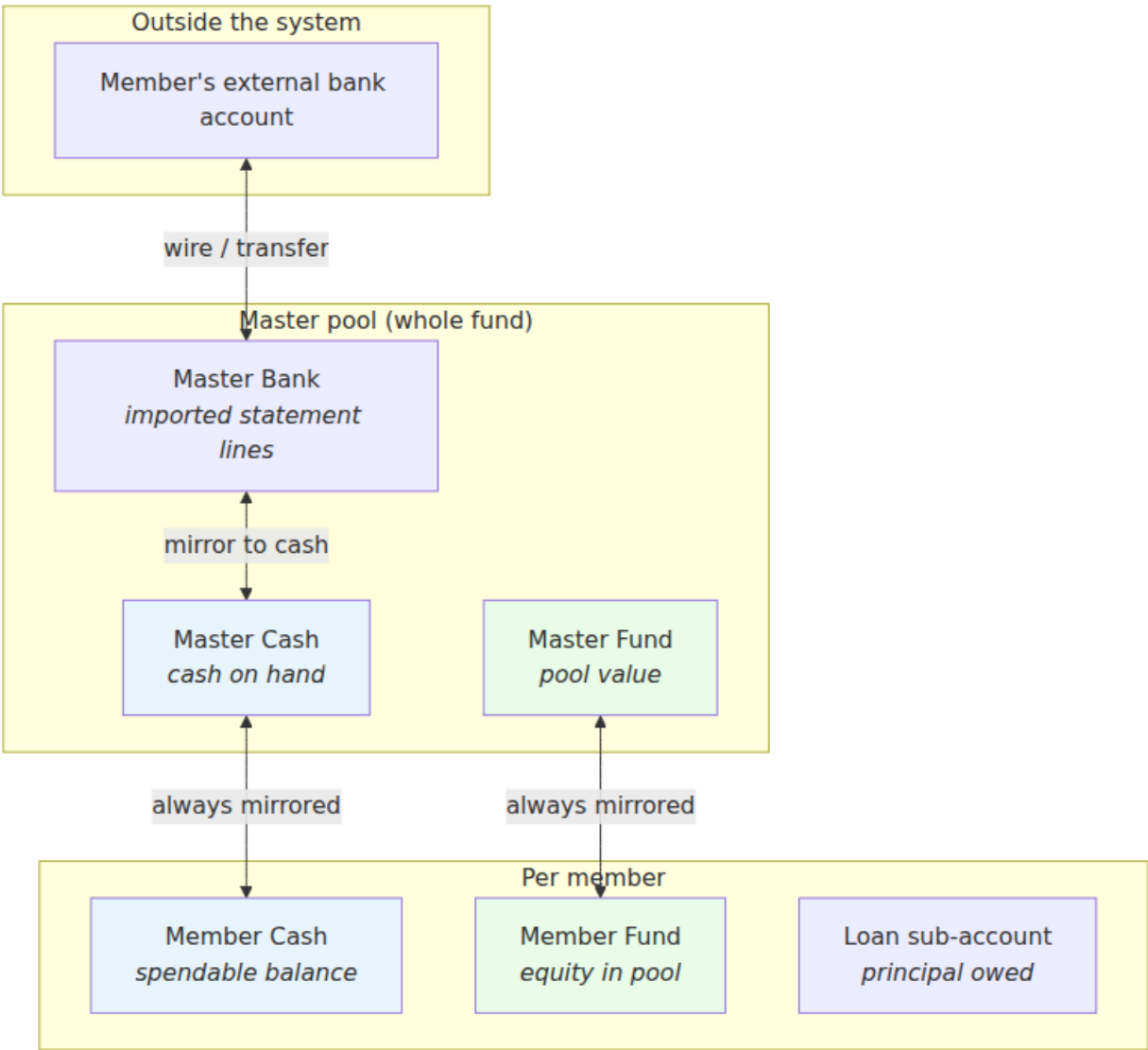


Figure 1: Diagram 1

Pool rules (checked nightly):

Invariant	Meaning
Master Cash = Σ Member Cash	Total cash in the pool equals the sum of all members' cash

Invariant	Meaning
Master Fund = Σ Member Fund	Total fund value equals the sum of all members' fund balances

Cash and fund are **separate ledgers**. A member can have cash without fund movement (and vice versa) until an operation links them (e.g. contribution collection moves cash $\rightarrow$ fund).

2. How money enters the system

Two main paths; both end with **member cash credited** (and master cash mirrored).

Important: Ledger posting records **intent** (money is in the member's cash account). **Bank clearance** (matching an imported line or an uncleared placeholder) confirms it against the real bank — it does **not** post extra cash/fund legs when matched correctly.

3. Monthly contribution collection

When a member has cash (from a deposit or legacy import mirror), the collection cycle **moves cash into fund equity**.

Economic meaning: Cash leaves the member's "wallet" and becomes **fund equity** (their share of the pool). Late fees, if any, also debit member + master cash.

4. Loan lifecycle — disbursement, cash-out, repayment

4a. Disbursement (ledger only — no bank payout yet)

4b. Cash-out (actual bank transfer)

4c. Loan repayment (EMI / imported payment)

Repayment target: Master fund slice + settlement threshold (e.g. 16%). The member's own fund portion at disbursement is **equity**, not cash EMI — only the pool's share (+ settlement) is collected via repayments.

5. Special case — 100% member-fund loan (e.g. loan #143)

When the full loan is funded from the member's own fund (`member_portion` = full amount, `master_portion` = 0, no settlement due):

There is **no pool exposure**, so no ongoing installment obligation.

6. End-to-end lifecycle (one member, simplified)

7. Mental model for stakeholders

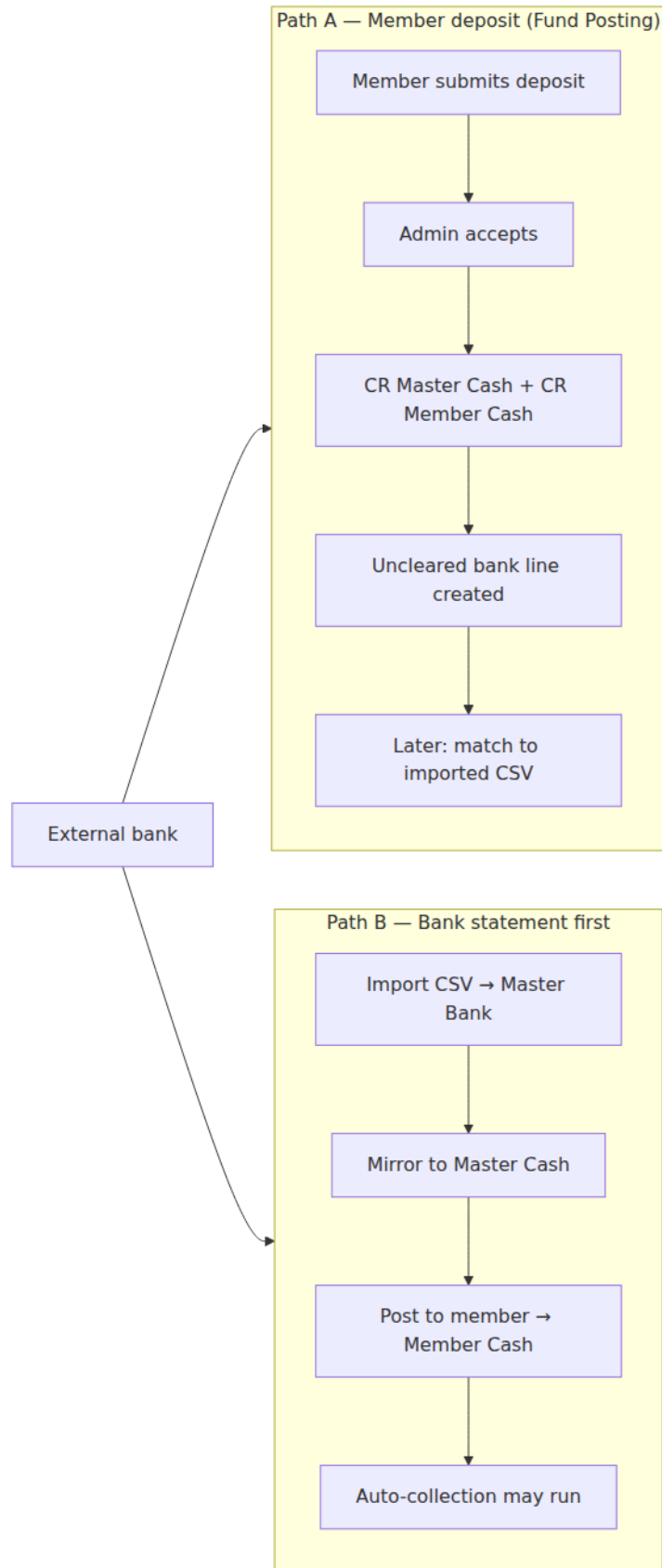


Figure 2: Diagram 2

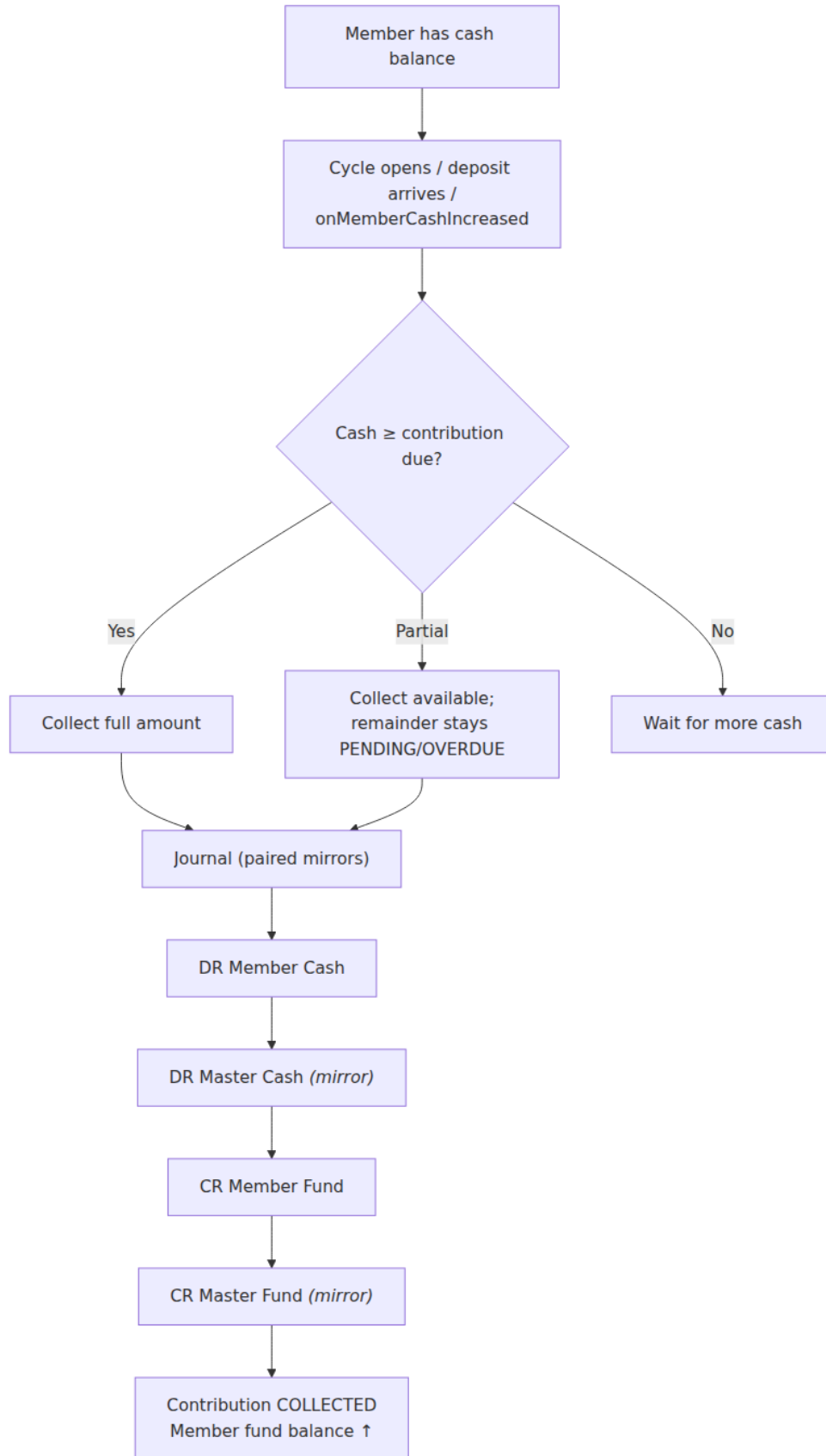


Figure 3: Diagram 3

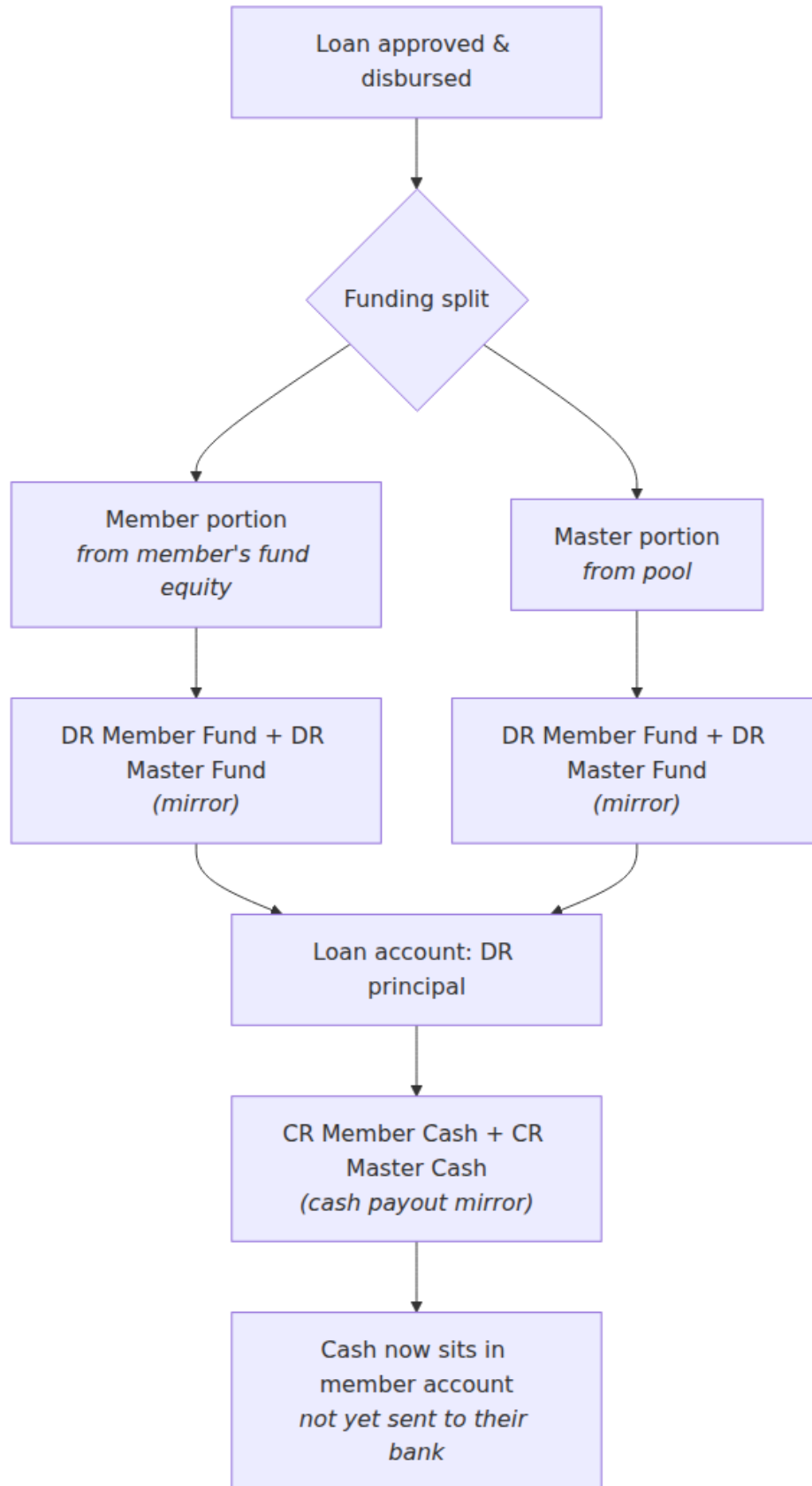


Figure 4: Diagram 4

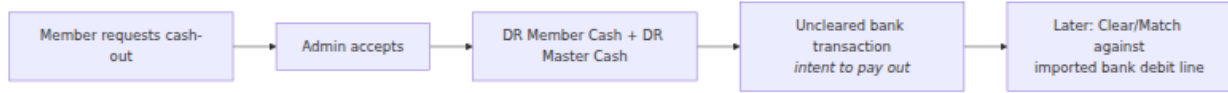


Figure 5: Diagram 5

Question	Answer
Where is “real” bank money?	Master Bank (imported statements) + uncleared lines until matched
What can a member spend?	Member Cash (after collection hasn’t taken it for contributions/EMIs)
What is their stake in the pool?	Member Fund (contributions – disbursements + repayments)
When does the pool shrink/grow?	Fund legs on contribution (in), disbursement (out), repayment (in)
Why two steps for bank?	Ledger = economic event; Clearance = proof it hit the bank

8. Operational functions → account touchpoints

Function	Cash legs	Fund legs	Bank
Deposit accept	CR member + master cash	—	Uncleared line → match
Bank import	Mirror → post to member	—	Master bank updated
Contribution	DR member + master cash	CR member + master fund	—
Loan disburse	CR member + master cash	DR member + master fund	—
Cash-out approve	DR member + master cash	—	Uncleared → match
Loan repayment	CR then DR cash (mirror pair)	CR member + master fund	—
Subscription fee	DR member + master cash	Fee to master fees	—
Legacy import	Same mirror patterns	Same mirror patterns	Often skipped

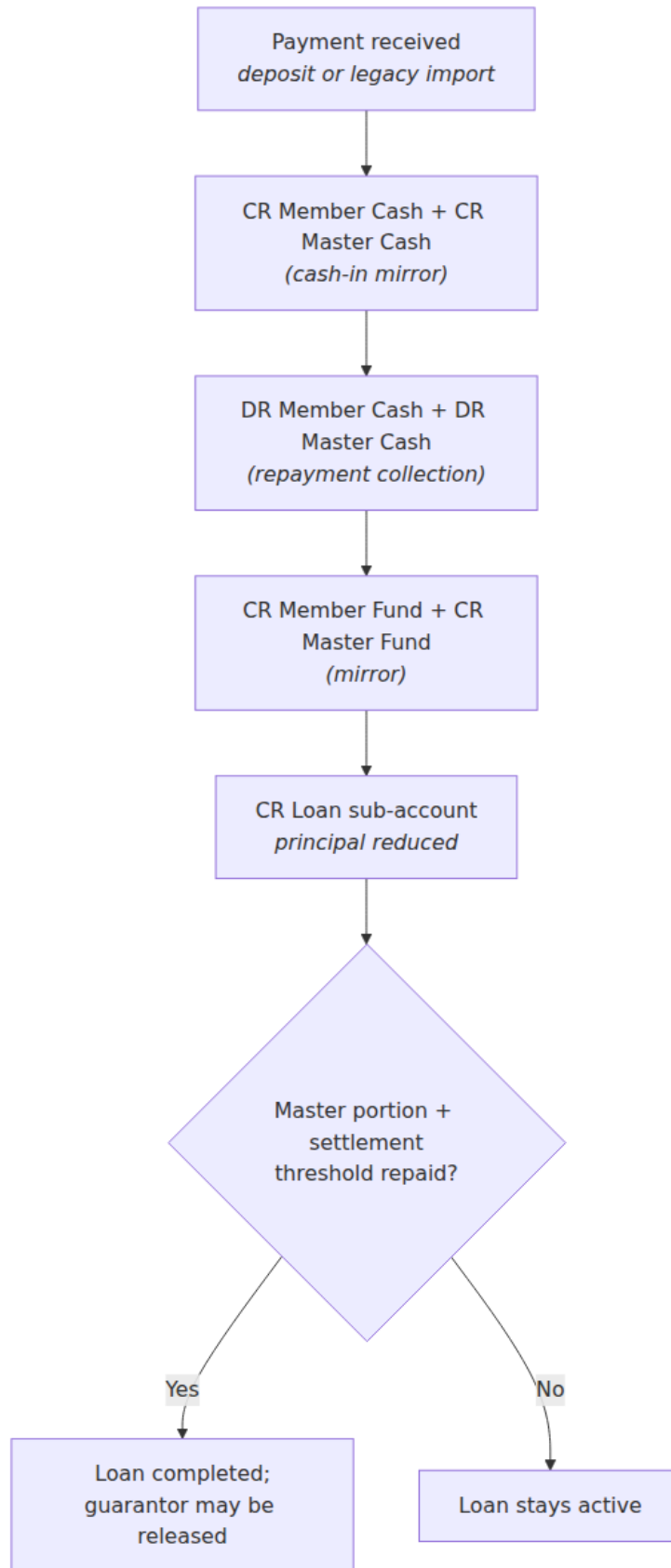


Figure 6: Diagram 6

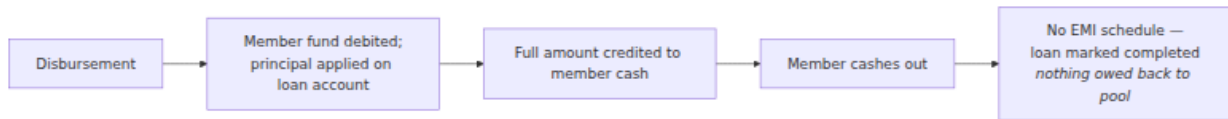


Figure 7: Diagram 7

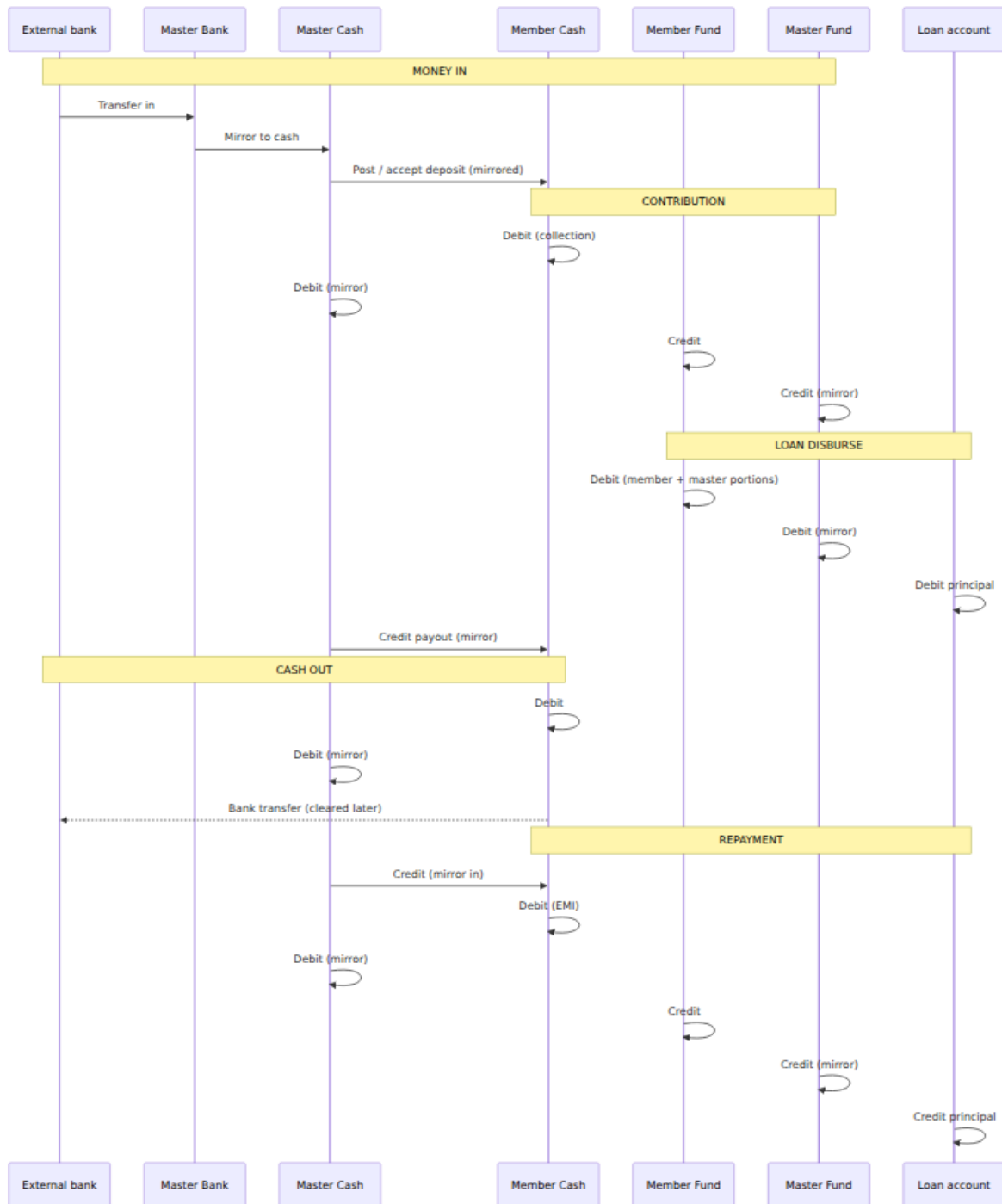


Figure 8: Diagram 8